

Statistics Homework 4:

Simple Regression

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Part A: Neuroticism and Marital Satisfaction

Bouchard, Lussier, and Sabourin (1999) investigated the association between relationship satisfaction and neuroticism in a sample of 446 married couples. Neuroticism is conceptualized as a personality trait related to a person's tendency to experience negative emotions. If a person is described as being high on neuroticism, it means the person gets upset easily and experiences more negative emotion than most other people. In this study, each partner in each couple completed a measure of neuroticism (from the NEO-PI) and a measure of relationship satisfaction (the Dyadic Adjustment Scale). Results for wives are printed below.

Wives neuroticism and marital satisfaction:

Inter-Item Covariance Matrix

	w_das	w_neuro
w_das	402.59	-79.15
w_neuro	-79.15	98.11

Note:

w_das = wife's marital satisfaction (Dyadic Adjustment Scale)

w_neuro = wife's neuroticism

Descriptive Statistics

	Mean	Std. Deviation	N
w_das	106.21	20.06	446
w_neuro	49.78	9.90	446

Correlations

		w_das	w_neuro
w_das	Pearson Correlation	1	-.40**
	Sig. (2-tailed)		.000
	N	446	446
w_neuro	Pearson Correlation	-.40**	1
	Sig. (2-tailed)	.000	
	N	446	446

** . Correlation is significant at the 0.01 level (2-tailed).

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.398 ^a	.159	.157	18.425

a. Predictors: (Constant), w_neuro

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	28417.6	1	28417.56	83.70	.000 ^a
	Residual	150737.2	444	339.50		
	Total	179154.7	445			

a. Predictors: (Constant), w_neuro

b. Dependent Variable: w_das

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	146.37	4.48		32.70	.000
w_neuro	-.81	.09	-.40	-9.15	.000

a. Dependent Variable: w_das

Questions

- In solving a regression equation, the computer attempts to make a certain value as small as possible. In the printed results, what number is this minimized value?
- What is the typical distance that people's scores on the outcome variable deviated from the mean?
- What is the typical distance that people's scores on the outcome variable deviated from the regression slope?
- If this study were repeated over and over, how much will the estimated (unstandardized) beta typically deviate from the true population parameter?
- If this study were repeated over and over, how much will the estimated correlation typically deviate from the true population parameter? (note: this requires a calculation)
- What is the percent of variance that has been explained in this equation?
- What is the expected increase in y if there is a one unit increase in x?
- What is the expected value for y when x equals zero?
- In the equation for variance in a sample ($y_{is} = b_{0s} + b_{1s}x_{is} + e_{is}$), what is the average deviation of $|e_{is}|$?
- In the equation for sampling error variance ($b_{1sp} = \beta_{1p} + e_{1sp}$), what is the average deviation of $|e_{1sp}|$?
- If the null hypothesis about the association between x and y is true, what is the standardized e_{1sp} for this study?
- What is $\Sigma(y - \bar{y})^2$?
- What is $\frac{\Sigma(y - \bar{y})^2}{n - 1}$?
- What is $\Sigma(x - \bar{x})(y - \bar{y})$?
- What is $\frac{\Sigma(x - \bar{x})(y - \bar{y})}{n - 1}$?
- If Genevieve received a score of 55 on neuroticism, what is Genevieve's predicted score for relationship satisfaction?
- If Genevieve actually received a score of 114 on relationship satisfaction, what is the squared error for Genevieve's score?
- According to the results printed above, what is the typical squared error for participants in this study?
- Was the squared error for Genevieve higher or lower than the typical squared error for participants in this study?
- According to the results printed above, what is the sum of squared errors for all participants in this study?

Is there an association between neuroticism and relationship satisfaction? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

21. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
22. the value of the descriptive statistic
23. the name of the inferential statistic (use the Acceptable Abbreviations Page)
24. the value of the inferential statistic
25. the p-value (**report exactly as it appears with no extra zeros and no rounding**)

26. The association is best described as:

- (A) not significant (B) significant but trivial (C) small (D) medium (E) large

Results regarding husbands' neuroticism and relationship satisfaction are printed below.

Inter-Item Covariance Matrix

	h_das	h_neuro
h_das		-43.07
h_neuro	-43.07	98.23

Note:

h_das = husband's marital satisfaction (Dyadic Adjustment Scale)

h_neuro = husband's neuroticism

Descriptive Statistics

	Mean	Std. Deviation	N
h_das	108.31		446
h_neuro	49.79		446

Correlations

		h_das	h_neuro
h_das	Pearson Correlation	1	
	Sig. (2-tailed)		
	N	446	446
h_neuro	Pearson Correlation		1
	Sig. (2-tailed)		
	N	446	446

**. Correlation is significant at the 0.01 level

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1				

a. Predictors: (Constant), h_neuro

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression					
	Residual	138584.6				
	Total	146988.9				

a. Predictors: (Constant), h_neuro

b. Dependent Variable: h_das

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)		4.29		30.34	.000
h_neuro		.08			

a. Dependent Variable: h_das

Compute the following:

27. variance for the dependent variable
28. standard deviation for the dependent variable
29. standard error of estimate
30. regression sum of squares
31. total degrees of freedom
32. degrees of freedom for “regression”
33. degrees of freedom for the residual
34. mean square regression
35. mean square residual (error)
36. F value
37. Significance of F
38. R squared
39. beta weight
40. intercept
41. standardized beta
42. *t* value for the beta weight
43. significance of *t* for the beta weight

Is there an association between neuroticism and relationship satisfaction? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the *Acceptable Abbreviations Page*.

44. the name of the descriptive statistic (use the *Acceptable Abbreviations Page*)
45. the value of the descriptive statistic
46. the name of the inferential statistic (use the *Acceptable Abbreviations Page*)
47. the value of the inferential statistic
48. the p-value
49. The association is best described as:
(A) not significant (B) significant but trivial (C) small (D) medium (E) large

Part B: Negative Communication and Relationship Satisfaction

In an early study investigating communication in couples, Julien, Markman, and Lindahl (1989) assessed negative communication and relationship satisfaction in a sample of 59 couples. Each couple was assessed on two occasions, once before they were married and a second time after they had been married for five years. In the first assessment, couples were video recorded discussing an area of relationship conflict. A team of research assistants then rated the extent of negative escalation in each conversation. At the second assessment session, after couples had been married five years, couples completed a measure of relationship satisfaction called the Marital Adjustment Test (and scores from both partners were averaged together). A key question addressed in this study is whether the amount of negative escalation before a couple marries is associated with relationship satisfaction five years after they marry. Results are printed below.

Inter-Item Covariance Matrix

	neg_escalation	MAT
neg_escalation	2.24	-8.42
MAT	-8.42	327.35

Note:

neg_escalation = couple's negative escalation

MAT = Marital Adjustment Test (relationship satisfaction)

Descriptive Statistics

	Mean	Std. Deviation	N
MAT	126.34	18.09	59
neg_escalation	2.38	1.50	59

Correlations

		neg_escalation	MAT
neg_escalation	Pearson Correlation	1	-.31*
	Sig. (2-tailed)		.017
	N	59	59
MAT	Pearson Correlation	-.31*	1
	Sig. (2-tailed)	.017	
	N	59	59

*. Correlation is significant at the 0.05 level (2-tailed).

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.311 ^a	.097	.081	17.347

a. Predictors: (Constant), neg_escalation

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1833.2	1	1833.23	6.09	.017 ^a
	Residual	17153.0	57	300.93		
	Total	18986.3	58			

a. Predictors: (Constant), neg_escalation

b. Dependent Variable: MAT

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	135.29	4.27		31.68	.000
	neg_escalation	-3.76	1.52	-.31	-2.47	.017

a. Dependent Variable: MAT

Questions

50. In solving a regression equation, the computer attempts to make a certain value as small as possible. In the printed results, what number is this minimized value?
51. What is the typical distance that people's scores on the outcome variable deviated from the mean?
52. What is the typical distance that people's scores on the outcome variable deviated from the regression slope?
53. If this study were repeated over and over, how much will the estimated (unstandardized) beta typically deviate from the true population parameter?
54. If this study were repeated over and over, how much will the estimated correlation typically deviate from the true population parameter? (note: this requires a calculation)

55. What is the percent of variance that has been explained in this equation?
56. What is the expected increase in y if there is a one unit increase in x ?
57. What is the expected value for y when x equals zero?
58. In the equation for variance in a sample ($y_{is}=b_{0s}+b_{1s}x_{is}+e_{is}$), what is the average deviation of $|e_{is}|$?
59. In the equation for sampling error variance ($b_{1sp}=\beta_{1p}+e_{1sp}$), what is the average deviation of $|e_{1sp}|$?
60. If the null hypothesis about the association between x and y is true, what is the standardized e_{1sp} for this study?
61. What is $\Sigma(y - \bar{y})^2$?
62. What is $\frac{\Sigma(y - \bar{y})^2}{n - 1}$?
63. What is $\Sigma(x - \bar{x})(y - \bar{y})$?
64. What is $\frac{\Sigma(x - \bar{x})(y - \bar{y})}{n - 1}$?
65. If Danielle received a score of 4 on negative escalation, what is Danielle's predicted score for relationship satisfaction?
66. If Danielle actually received a score of 100 on relationship satisfaction, what is the squared error for Danielle's score?
67. According to the results printed above, what is the typical squared error for participants in this study?
68. Was the squared error for Danielle higher or lower than the typical squared error for participants in this study?
69. According to the results printed above, what is the sum of squared errors for all participants in this study?

Is there an association between negative escalation and relationship satisfaction? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

70. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
71. the value of the descriptive statistic
72. the name of the inferential statistic (use the Acceptable Abbreviations Page)
73. the value of the inferential statistic
74. the p-value (**report exactly as it appears with no extra zeros and no rounding**)
75. The association is best described as:
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

Part C: More Negative Communication and Relationship Satisfaction

Sanford (2003) also conducted a study investigating the relationship between negative communication and relationship satisfaction (actually, this was part of this writer's dissertation). In this study, 37 couples completed a series of video recorded problem solving discussions in a laboratory, and a team of research assistants rated the level of negative communication. Four years after the initial assessment, couples were contacted and asked to complete a relationship satisfaction questionnaire and return it via mail. From the original sample, 26 of the couples that were still intact returned the questionnaire. Satisfaction scores from each partner were averaged together to create a single satisfaction score for each couple. Results from this study are printed below.

Negative communication and relationship satisfaction

Inter-Item Covariance Matrix

	T2_sat	T1_neg
T2_sat		-.92
T1_neg	-.92	.20

Note:

T1_neg = Time 1 negative communication

T2_sat = Time 2 satisfaction

Descriptive Statistics

	Mean	Std. Deviation	N
T2_sat	-1.37		26
T1_neg	.83		26

Correlations

		T2_sat	T1_neg
T2_sat	Pearson Correlation	1	
	Sig. (2-tailed)		
	N	26	26
T1_neg	Pearson Correlation		1
	Sig. (2-tailed)		
	N	26	26

**. Correlation is significant at the 0.01 level (2-tailed).

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1				

a. Predictors: (Constant), T1_neg

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression					
	Residual	171.19				
	Total	277.28				

a. Predictors: (Constant), T1_neg

b. Dependent Variable: T2_sat

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)		1.12		2.19	.039
	T1_neg		1.20			

a. Dependent Variable: T2_sat

Compute the following:

76. variance for the dependent variable
77. standard deviation for the dependent variable
78. standard error of estimate
79. regression sum of squares
80. total degrees of freedom
81. degrees of freedom for “regression”
82. degrees of freedom for the residual
83. mean square regression
84. mean square residual (error)
85. F value
86. Significance of F
87. R squared
88. beta weight
89. intercept
90. standardized beta
91. t value for the beta weight
92. significance of t for the beta weight

Is there an association between negative communication and relationship satisfaction? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

93. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
 94. the value of the descriptive statistic
 95. the name of the inferential statistic (use the Acceptable Abbreviations Page)
 96. the value of the inferential statistic
 97. the p-value
98. The association is best described as:
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

Part D: Computing predicted and residual scores

Five people rated the extent to which they liked okra (x) and the extent to which they liked owls (y). A regression analysis was conducted, and it produced the following results.

$$b_0 = 3.4$$

$$b_1 = 0.3$$

$$\text{mean of } y = 5.5$$

$$sd \text{ of } y = 2.06$$

The scores on the x variable are listed below.

	x Likes Okra
Cal	1
Deb	5
Eve	7
Fay	9
Gus	13

Based on this information calculate:

99. The predicted owl liking score for Cal
100. The predicted owl liking score for Deb
101. The predicted owl liking score for Eve
102. The predicted owl liking score for Fay
103. The predicted owl liking score for Gus

104. The total sum of squares for the explained variance
105. The percent of variance explained

Five people rated the extent to which they liked kale (x) and the extent to which they liked koalas (y). A regression analysis was conducted, and it produced the following results.

$b_0 = 0.6$
 $b_1 = 1.2$
 mean of $y = 9.0$
 sd of $y = 4.47$

Listed below are scores for x , y , and predicted y .

	x Likes kale	y Likes koalas	Predicted y
Cal	4	7	5.4
Deb	6	3	7.8
Eve	7	9	9
Fay	8	15	10.2
Gus	10	11	12.6

Based on this information calculate:

106. The residual score for Cal
 107. The residual score for Deb
 108. The residual score for Eve
 109. The residual score for Fay
 110. The residual score for Gus

 111. What is the correlation between the residual scores you just created and the scores on x (likes kale)?
 112. What is the sum of squares for the error variance?
 113. What is the percent of variance explained?
-

Continued. . .

Five people rated the extent to which they liked figs (x) and the extent to which they liked foxes (y). A analysis with this dataset produced the following results.

$$r_{xy} = 1.0$$

$$\text{mean of } y = 7$$

$$sd \text{ of } y = 2.92$$

The scores on the y variable are listed below.

	y Likes Foxes
Cal	3
Deb	9
Eve	5
Fay	?
Gus	?

Based on this information calculate:

114. The predicted fox liking score for Cal

115. The predicted fox liking score for Deb

116. The predicted fox liking score for Eve

117. The residual score for Cal

118. The residual score for Deb

119. The residual score for Eve

120. What is the average of the residual scores for all five participants?

121. What is the standard deviation of the residual scores for all five participants?

Five people rated the extent to which they liked limes (x) and the extent to which they liked lemurs (y). A analysis with this dataset produced the following results.

$$r_{xy} = 0.0$$

$$\text{mean of } y = 7$$

$$sd \text{ of } y = 2.92$$

Some of the scores on the y variable are listed below.

	y Likes Lemurs
Cal	6
Deb	2
Eve	9
Fay	?
Gus	?

Based on this information calculate:

- 122. The predicted lemur liking score for Cal
- 123. The predicted lemur liking score for Deb
- 124. The predicted lemur liking score for Eve

- 125. The residual score for Cal
- 126. The residual score for Deb
- 127. The residual score for Eve

- 128. What is the average of the residual scores for all five participants?
- 129. What is the standard deviation of the residual scores for all five participants?

References

- Bouchard, G., Lussier, Y., & Sabourin, S. (1999). Personality and marital adjustment: Utility of the five-factor model of personality. *Journal of Marriage and the Family*, 61, 651-660.
- Julien, D., Markman, H. J., & Lindahl, K. M. (1989). A comparison of a global and a microanalytic coding system: Implications for future trends in studying interactions. *Behavioral Assessment*, 11, 81-100.
- Sanford, K. (2003). Problem-solving conversations in marriage: Does it matter what topics couples discuss. *Personal Relationships*, 10, 97-112.